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JOBSHEET 6

SORTING

```
package jobsheet5.Praktikum05;

import java.util.Scanner;

public class DosenMain {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        DataDosen list = new DataDosen();
        list.dataDosen = new Dosen[5];
        int menu;
        do {
            System.out.println("===Menu===");
            System.out.println("1. tambah data");
            System.out.println("2. tampil data");
            System.out.println("3. sorting asc");
            System.out.println("4. sorting dsc");
            System.out.print("masukan menu: ");
            menu = sc.nextInt();
            sc.nextLine();
            switch (menu) {
                case 1:
                    System.out.print("Kode : ");
                    String kode = sc.nextLine();

                    System.out.print("Nama : ");
                    String nama = sc.nextLine();

                    System.out.print("Jenis kelamin (T/F) : ");
                    boolean jenisKelamin = sc.nextBoolean();

                    System.out.print("Usia : ");
                    int usia = sc.nextInt();
                    sc.nextLine();

                    Dosen dsn = new Dosen(kode, nama, jenisKelamin, usia);
                    list.tambah(dsn);
```

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        break;
    case 2:
        list.tampil();
        break;
    case 3:
        list.SortingASC();
        list.tampil();
        break;
    case 4:
        list.SortingDSC();
        list.tampil();
        break;
    case 5:
        System.out.println("program selesai");
        break;
    default:
        System.out.println("Pilihan tidak valid");
    }
} while (menu != 5);
sc.close();
}
}

```

```

package jobsheet5.Praktikum05;

public class DataDosen {
    Dosen[] dataDosen = new Dosen[10];
    int idx;

    void tambah(Dosen dsn){
        if (idx < dataDosen.length) {
            dataDosen[idx] = dsn;
            idx++;
        } else {
            System.out.println("Data Sudah Penuh");
        }
    }

    void tampil() {
        if (showEmptyData()) {
            System.out.println("Data Dosen");
        }
    }
}

```

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        for (int i = 0; i < idx; i++) {
            dataDosen[i].tampil();
            System.out.println("-----");
        }
    }

}

boolean showEmptyData() {
    if (idx == 0) {
        System.out.println("Data dosen kosong");
        return false;
    } else {
        return true;
    }
}

void SortingASC() {
    if (showEmptyData()) {
        for (int i = 1; i < dataDosen.length; i++) {
            Dosen temp = dataDosen[i];
            int j = i;
            while (j > 0 && dataDosen[j-1].usia > temp.usia) {
                dataDosen[j] = dataDosen[j-1];
                j--;
            }
            dataDosen[j] = temp;
        }
    }
}

void SortingDSC() {
    if (showEmptyData()) {
        for (int i = 1; i < dataDosen.length; i++) {
            Dosen temp = dataDosen[i];
            int j = i;
            while (j > 0 && dataDosen[j-1].usia < temp.usia) {
                dataDosen[j] = dataDosen[j-1];
                j--;
            }
            dataDosen[j] = temp;
        }
    }
}

```

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void InsertionSort() {
    if (showEmptyData()) {
        for (int i = 1; i < dataDosen.length; i++) {
            Dosen temp = dataDosen[i];
            int j = i;
            while (j > 0 && dataDosen[j-1].usia > temp.usia) {
                dataDosen[j] = dataDosen[j-1];
                j--;
            }
            dataDosen[j] = temp;
        }
    }
}

```

```

package jobsheet5.Praktikum05;

public class Dosen {
    String kode, nama;
    boolean jenisKelamin;
    int usia;

    Dosen() {}
    Dosen(String kd, String name, boolean jk, int age) {
        kode = kd;
        nama = name;
        jenisKelamin = jk;
        usia = age;
    }

    void tampil() {
        System.out.println("nama: " + nama);
        System.out.println("kode: " + kode);
        System.out.println("Jenis Kelamin : " + (jenisKelamin ? "Pria" :
"Wanita"));
        System.out.println("usia: " + usia);
    }
}

```

